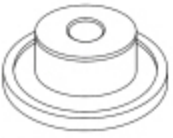
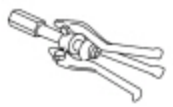


Transfer Case — 2-Speed Torque-On-Demand



Special Tool(s)

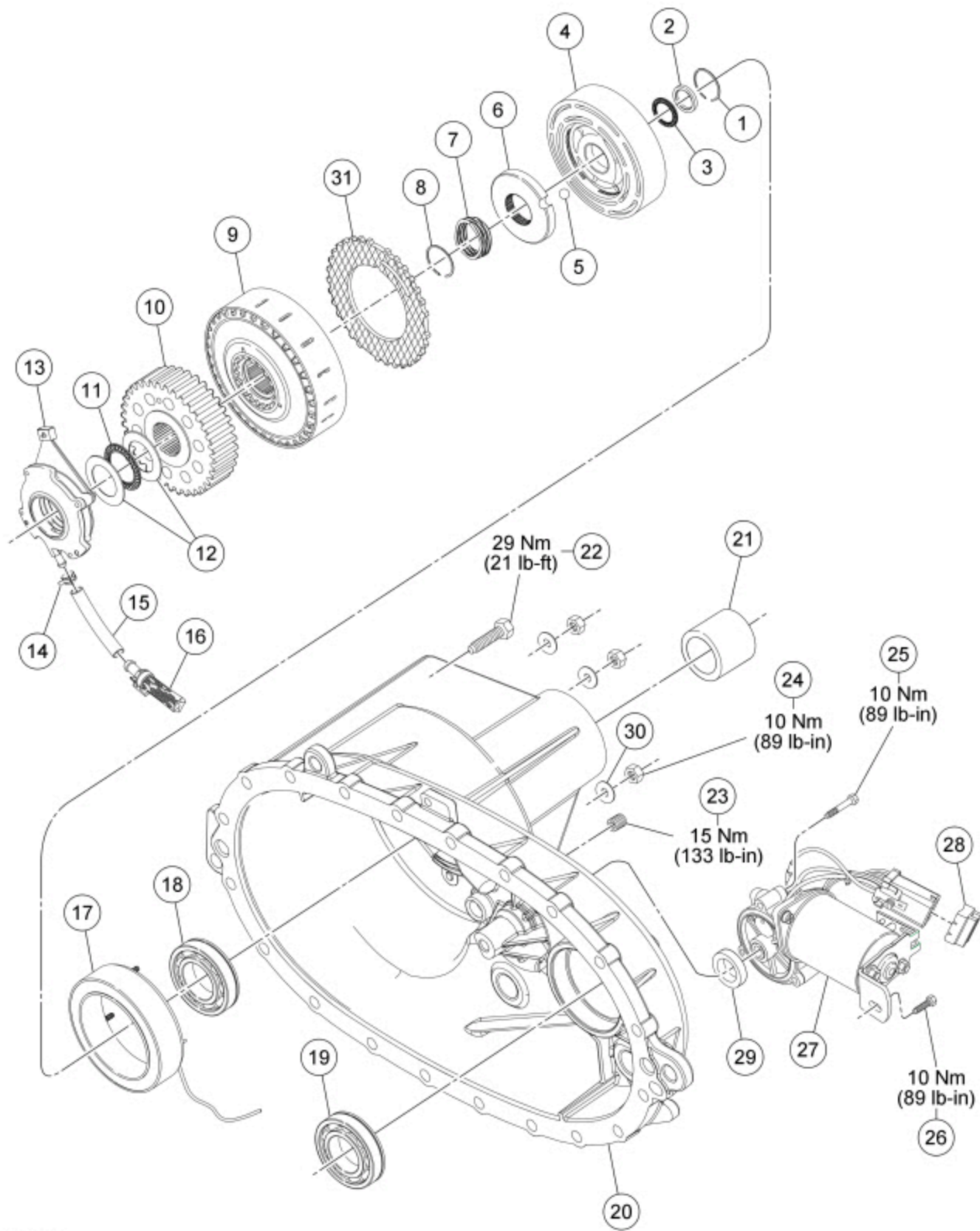
 ST1255-A	Handle 205-D055 (D81L-4000-A) or equivalent
 ST1186-A	Holding Fixture, Transmission 307-003 (T57L-500-B)
 ST1786-A	Installer, Front Axle Oil Seal 205-350 (T95T-3010-A)
 ST1789-A	Installer, Input Shaft Bearing 308-085 (T83T-7025-C)
 ST2305-A	Installer, Input Shaft Oil Seal 308-186 (T90T-7127-B)
 ST2261-A	Installer, Input Shaft Oil Seal 308-249 (T96T-7127-A)
 ST1788-A	Installer, Mainshaft Bearing 308-060 (T77J-7025-K)

 ST2239-A	Installer, Output Shaft Front Bearing 308-410
 ST1466-A	Installer, Valve Stem Oil Seal 303-367 (T90P-6510-AH)
 ST1200-A	Remover, Bearing Cup 308-047 (T77F-1102-A)
 ST1185-A	Slide Hammer 100-001 (T50T-100-A)

Material

Item	Specification
Motorcraft® MERCON® LV Automatic Transmission Fluid XT-10-QLVC (US); CXT-10-LV12 (Canada)	MERCON® LV
Ultra Silicone Sealant TA-29	—

2-Speed Transfer Case — Rear Half

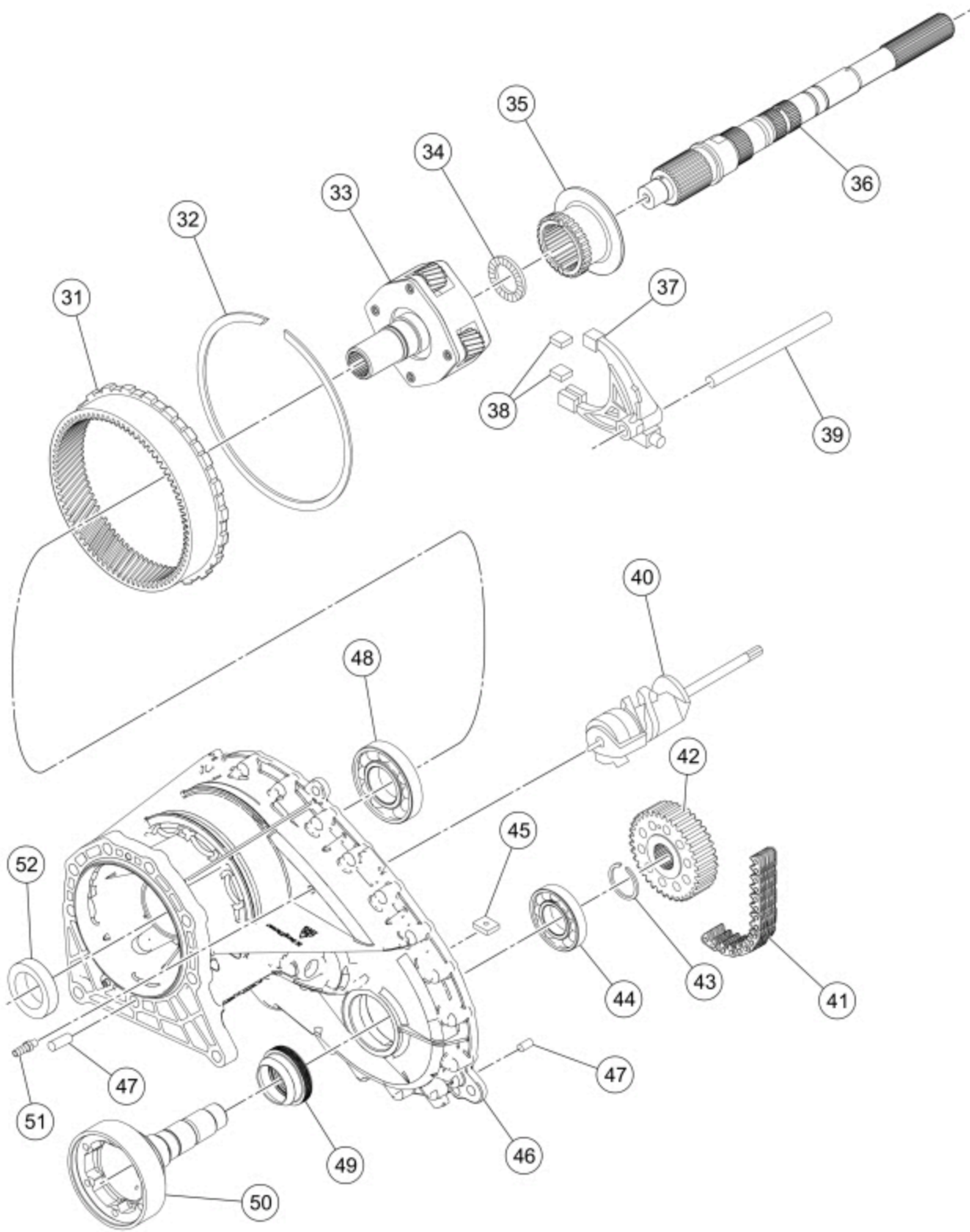


N0130853

Item	Part Number	Description
1	79177917	Snap ring
2	7N0637N063	Spacer
3	7D2217D221	Caged thrust bearing
4	7G3627G362	Hub and coil housing
5	7A5347A534	Ball (3 required)
6	7R4287R428	Cam
7	7D1267D126	Wave spring

8	7917 7917	Snap ring
9	7C108 7C108	Clutch assembly
10	7177 7177	Drive sprocket
11	7D221 7D221	Caged thrust bearing
12	7C341 7C341	Thrust washer (2 required)
13	7A149 7A149	Oil pump assembly
14	7H298 7H298	Hose clamp
15	4090 4090	Oil pump pickup hose
16	7A098 7A098	Oil pickup and filter
17	7G361 7G361	Transfer case field coil
18	7025 7025	Rear output shaft support bearing
19	7025 7025	Front output shaft rear bearing
20	7005 7005	Case
21	—	Output shaft bushing (part of 7005)
22	7A443 7A443	Transfer case bolts (20 required)
23	7A010 7A010	Fill plug
24	7045 7045	Coil nut (3 required)
25	N800670 N800670 -S-S	Shift motor bolt (3 required)
26	7A443C 7A443C	Shift motor bracket bolt
27	7G360 7G360	Transfer case shift motor
28	7H298 7H298	Inner connector retainer
29	7B215 7B215	Shift motor seal
30	7B368 7B368	Washer (3 required)
31	7H150 7H150	Armature

2-Speed Transfer Case — Front Half



N0161889

Item	Part Number	Description
31	7A1537A153	Ring gear
32	70647064	Snap ring
33	7A3987A398	Front planet carrier
34	7D2217D221	Front planet needle bearing
35	7D1647D164	Reduction hub
36	70617061	Output shaft
37	73897389	Shift fork

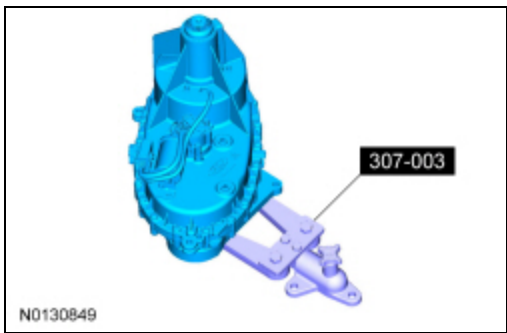
38	7C4307C430	Shift fork pad (2 required)
39	72407240	Shift fork shaft
40	7F0637F063	Shift cam assembly
41	7A0297A029	Drive chain
42	71777177	Driven sprocket
43	79177917	Snap ring
44	70257025	Front output shaft support bearing
45	7E2907E290	Magnet
46	70057005	Case
47	7B3627B362	Dowel
48	70257025	Front input shaft support bearing
49	7B2157B215	Front output shaft oil seal
50	70617061	Front output shaft
51	70347034	Vent tube
52	7B2157B215	Front input shaft seal

Disassembly

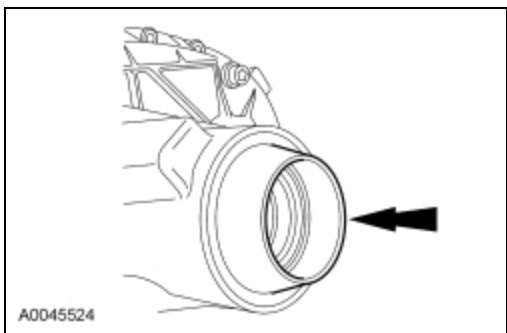
- 1. NOTICE: Seal all openings before cleaning the transfer case exterior. This will prevent entry of dirt and water, which may cause damage to internal components.**

Clean the transfer case exterior with solvent and dry with compressed air.

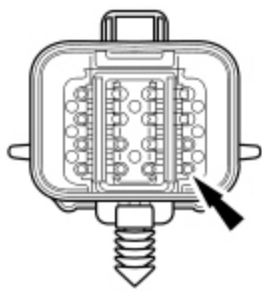
2. Using the Transmission Holding Fixture, secure the transfer case to a work bench. Rotate the transfer case so that the output shaft is pointing upward.



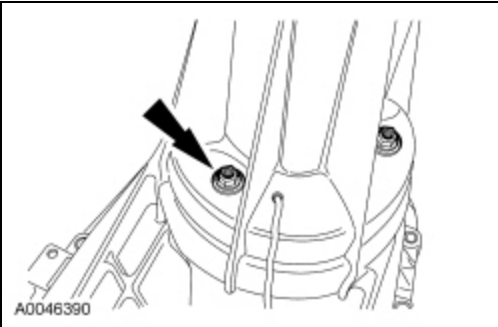
3. Remove and discard the rear output shaft seal.



4. Remove the coil wire pin from the electrical connector.
 - Remove the wire connector inner retainer.
 - Press the release tab and remove the coil wire pin (tan wire).

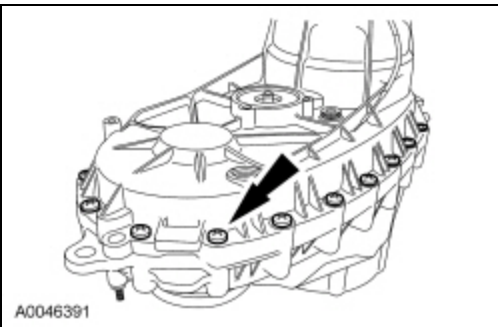


5. Loosen the 2 transfer case shift motor bracket nuts and remove the bracket bolt.
6. Remove the 3 transfer case shift motor bolts and transfer case shift motor.
7. Remove the 3 coil nuts and washers.

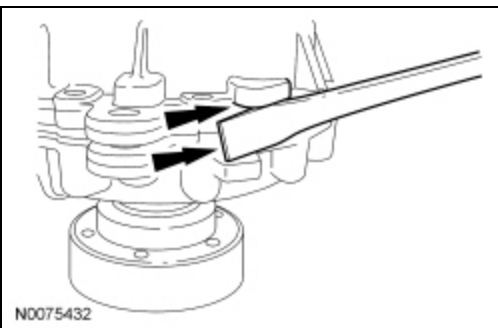


8. **NOTE:** *It is normal to find metal shavings after removing the self-tapping transfer case bolts.*

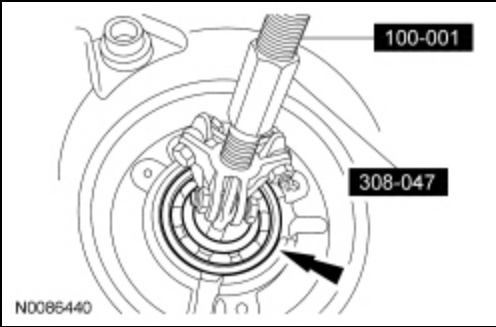
Remove the 20 transfer case cover bolts.



9. Using the pry bosses, separate the transfer case halves.

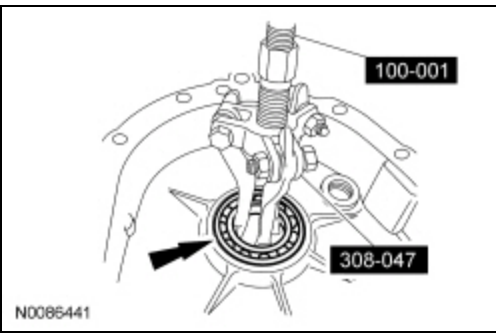


10. Using the Bearing Cup Remover and the Slide Hammer, remove the rear output shaft support bearing.
 - Inspect the bearing for wear or damage. Install a new bearing as necessary.



11. Using the Bearing Cup Remover and the Slide Hammer, remove the front output shaft rear bearing.

- Inspect the bearing for wear or damage. Install a new bearing as necessary.



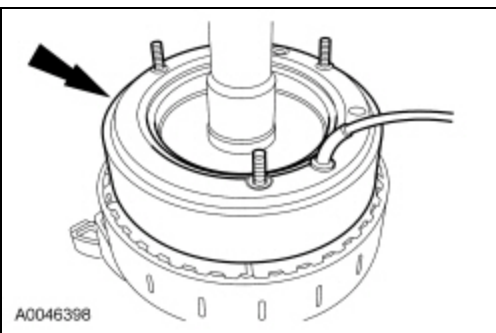
12. Remove the snap ring.

- Inspect the snap ring for wear or damage. Install a new snap ring as necessary.

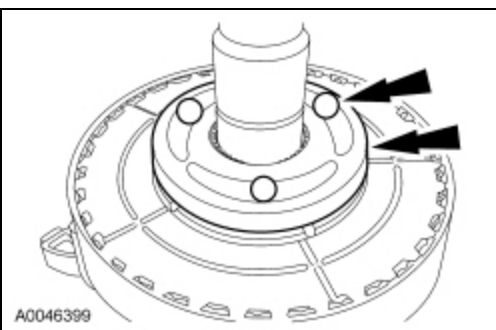
13. Remove the caged thrust bearing assembly.

- Inspect the bearing for wear or damage. Install a new bearing as necessary.

14. Remove the cam/coil housing.

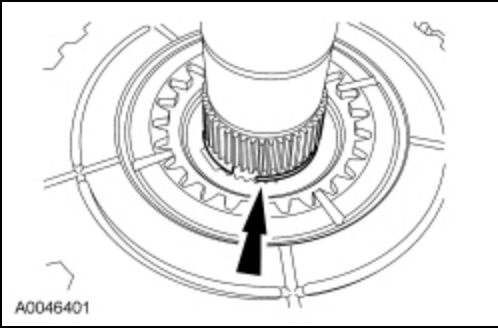


15. Remove the steel balls and the cam.



16. Remove the wave spring.

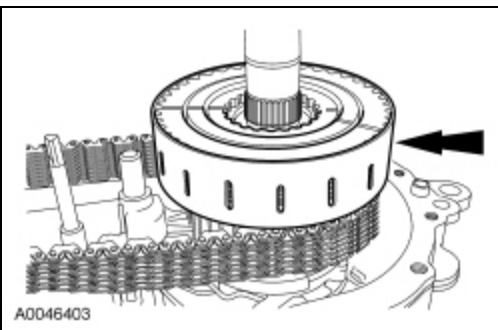
17. Remove and discard the snap ring.



18. **NOTICE:** When removing the clutch pack assembly, do not separate the clutch pack assembly. Keep tension on the clutch pack during removal. Set the clutch pack assembly on the bench in the same position as located in the transfer case. The thrust washer in the lower clutch pack uses tabs to hold it in place. If the thrust washer is not in place, a transfer case clearance problem will occur.

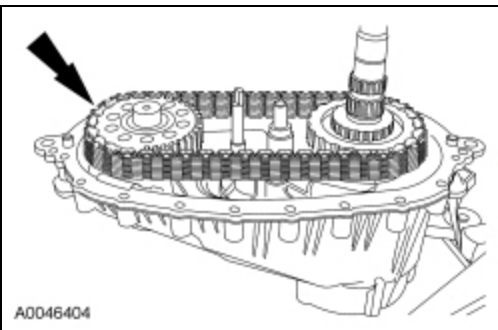
NOTICE: If the clutch pack is disassembled, a new clutch pack must be installed and the armature transferred. Failure to install a new clutch pack can result in component failure.

Remove the clutch pack assembly with the armature.



19. Remove the drive chain and the sprockets as an assembly.

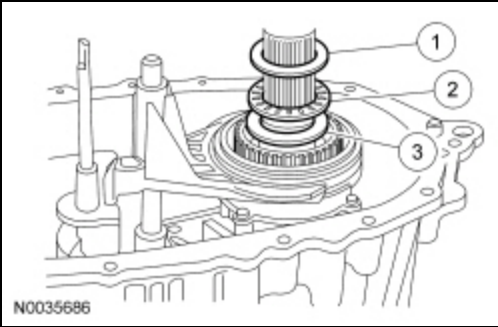
- Inspect the drive chain for wear or damage. Inspect the sprockets for broken or cracked teeth. Install new components as necessary.



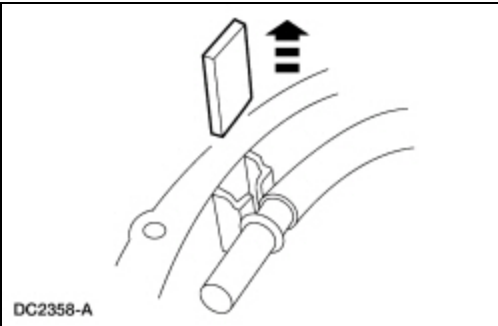
20. **NOTE:** The thrust washers are beveled on one side. Note orientation of the thrust washers before removal.

Remove the thrust bearing assembly.

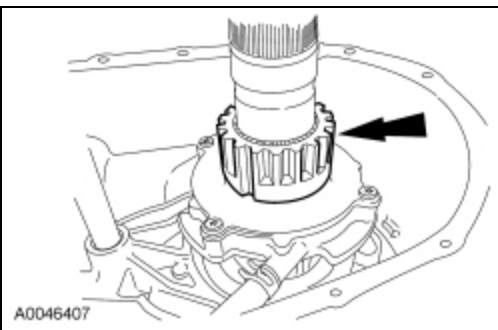
1. Remove the thrust washer.
 - The upper thrust washer may be attached to the sprocket.
2. Remove the thrust bearing.
3. Remove the thrust washer.
 - Inspect the washers and bearing for wear or damage. Install new components as necessary.



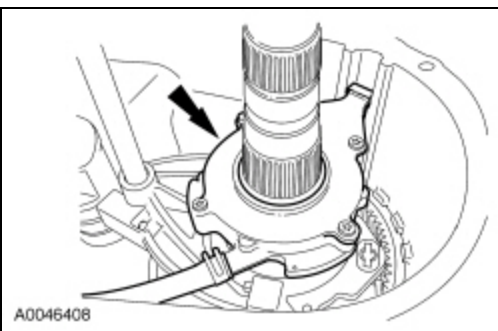
21. Remove and clean the oil pan magnet.



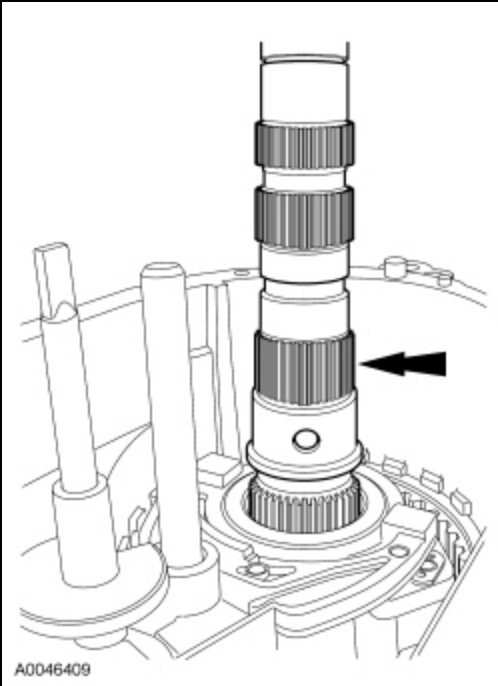
22. Remove the output shaft sprocket.



23. Remove the oil pump assembly.

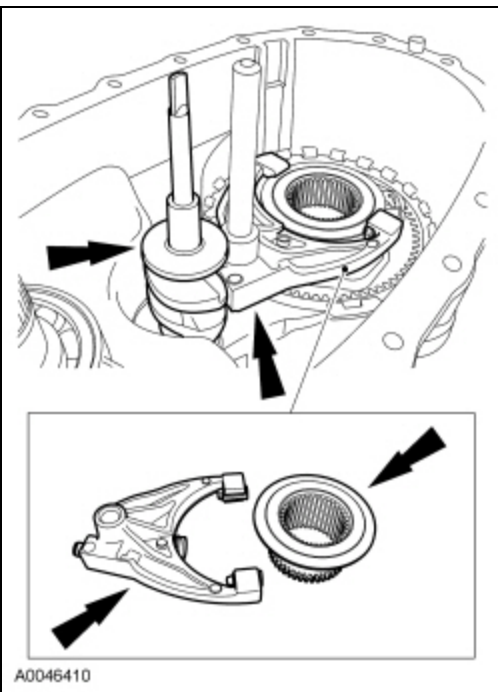


24. Remove the output shaft.

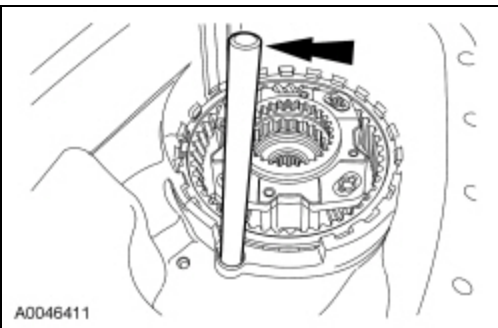


25. Remove the reduction shift fork, the reduction hub and the shift cam as an assembly.

- Separate and inspect the reduction hub and the reduction shift fork for wear or damage. Do not disassemble the shift cam assembly.

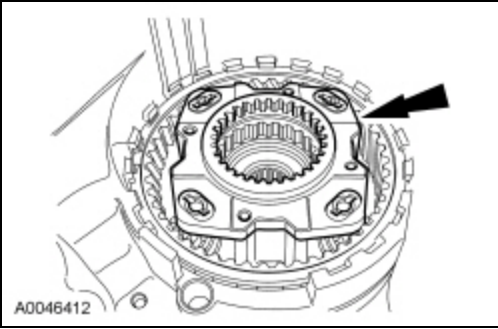


26. Remove the shift fork shaft.



27. Remove the front planet.

- Remove the front planet needle bearing, if necessary.

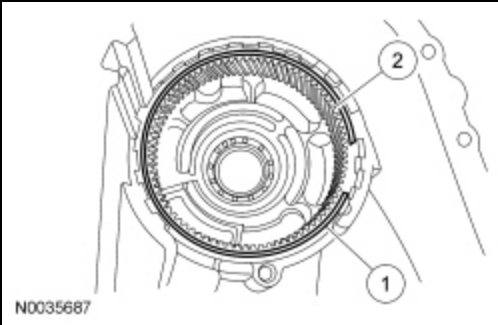


28. Remove the ring gear.

1. Remove the snap ring.

- Inspect the snap ring for wear or damage. Install a new snap ring as necessary.

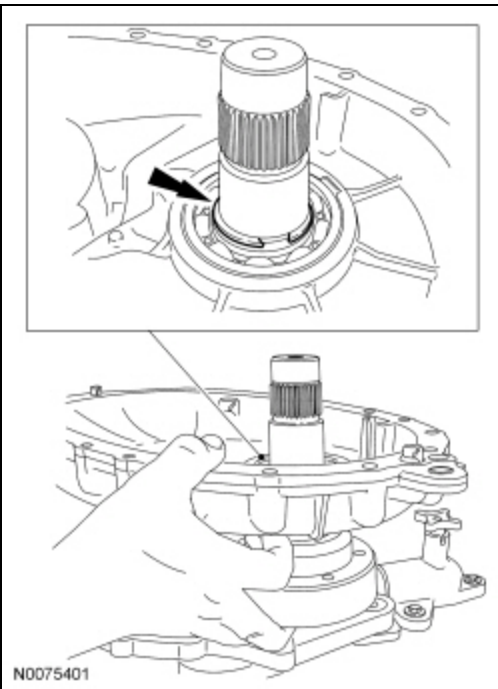
2. Remove the ring gear.



29. **NOTE:** *Do not let the front output shaft fall when removing the snap ring.*

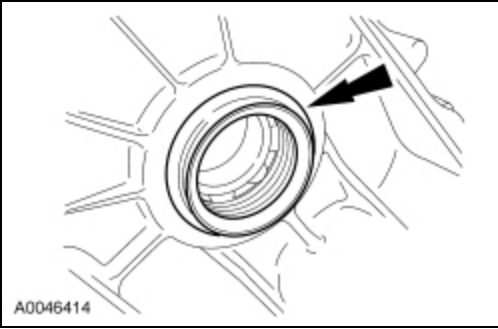
Remove the snap ring and the front output shaft.

- Inspect the snap ring for wear or damage. Install a new snap ring as necessary.



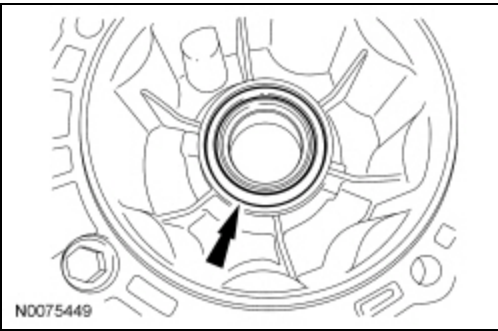
30. **NOTICE:** *Carefully remove the seal, do not damage the seal bore.*

Remove and discard the front output shaft oil seal.

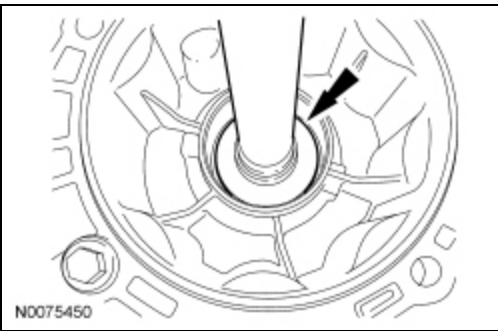


31. **NOTICE:** Carefully remove the seal, do not damage the seal bore.

Remove and discard the front input shaft seal.

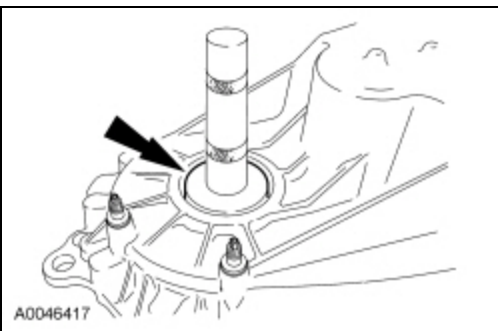


32. Remove the front planet support bearing.

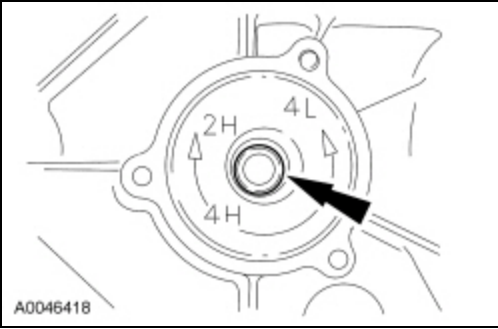


33. Remove the front output shaft support bearing.

- Inspect the bearing for wear or damage. Install a new bearing as necessary.

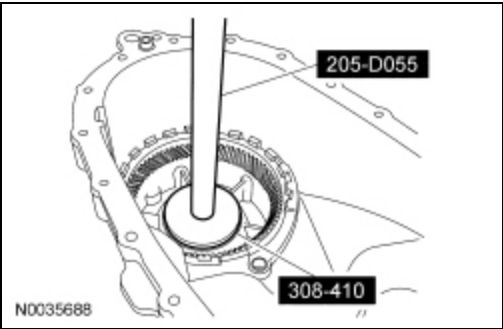


34. Remove and discard the shift shaft seal.

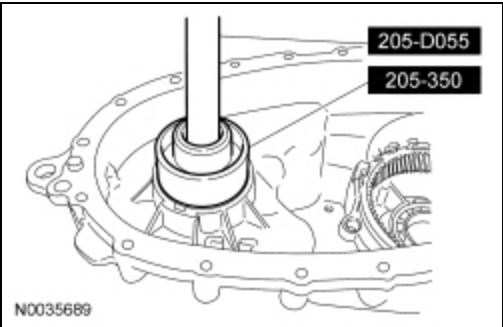


Assembly

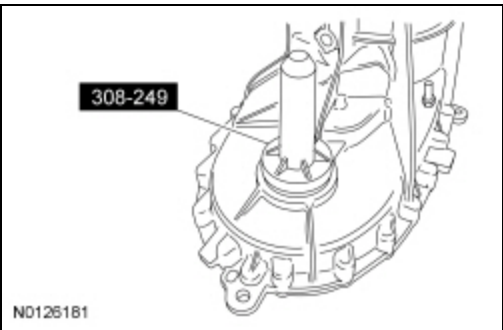
1. Lubricate all internal components with the recommended transfer case lubricant during reassembly.
2. Clean the mating surfaces of the cases.
3. Using a press and the Output Shaft Front Bearing Installer with the Handle, install the front planet support bearing.



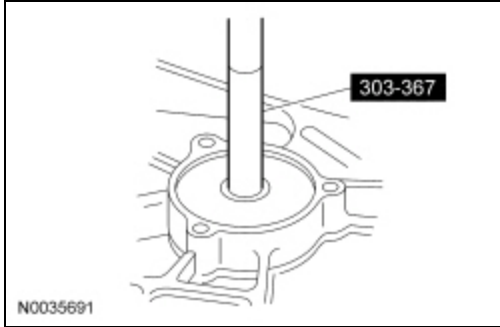
4. Using a press and the Front Axle Oil Seal Installer with the Handle, install the front output shaft support bearing.



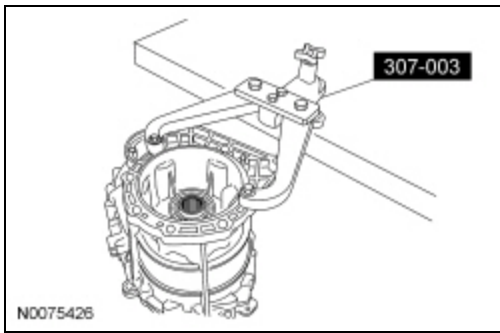
5. Using the Input Shaft Oil Seal Installer, install a new front output shaft seal.



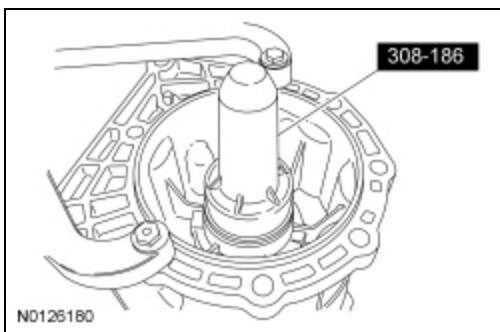
6. Using the Valve Stem Oil Seal Installer, install a new shift shaft seal.



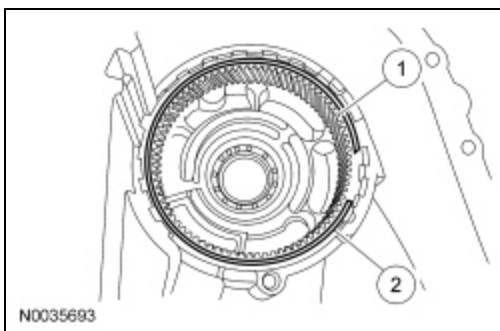
7. Install the case to the Transmission Holding Fixture. Secure the case and the Transmission Holding Fixture to the work bench.



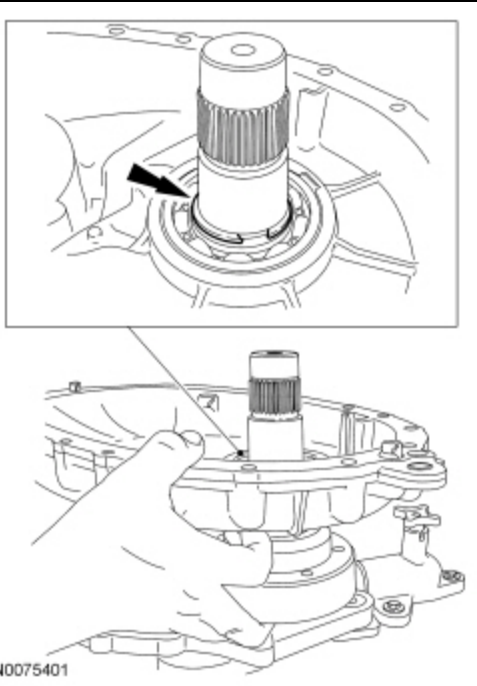
8. Using Input Shaft Oil Seal Installer, install a new front input shaft seal.



9. Install the ring gear.
 1. Position the ring gear in the case.
 2. Install the snap ring.

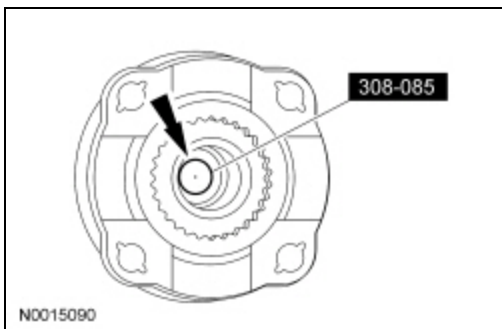


10. Install the front output shaft and the snap ring.

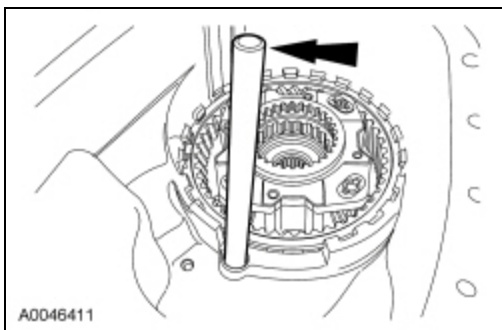


11. **NOTICE:** If removed, using a press and the Input Shaft Bearing Installer, install the needle bearing into the front planet. Be careful not to damage the bearing cage.

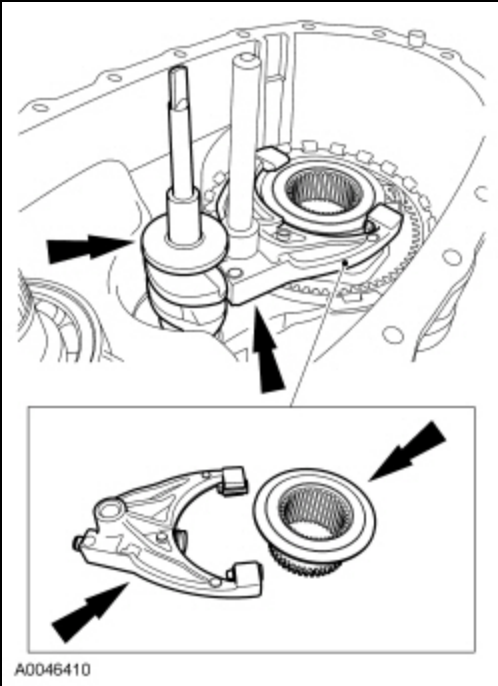
Install the front planet.



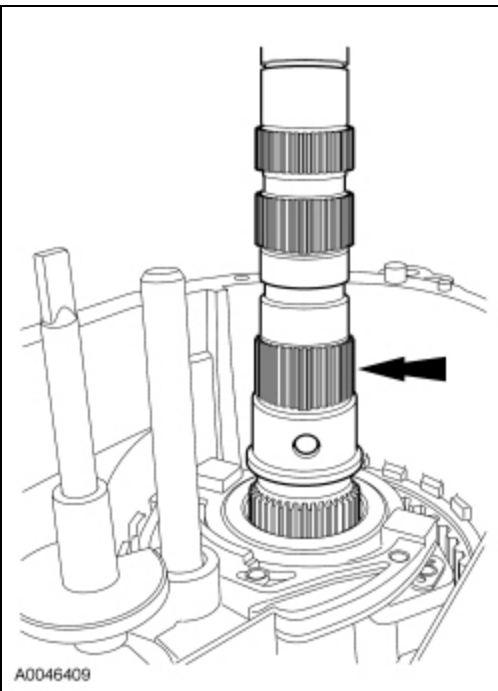
12. Install the shift fork shaft.



13. Assemble the reduction hub and the reduction shift fork. Install the reduction shift fork, the reduction hub and the shift cam as an assembly.

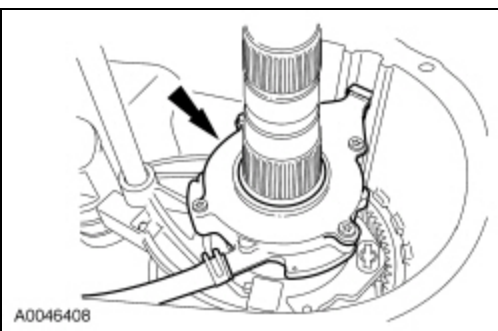


14. Install the output shaft.

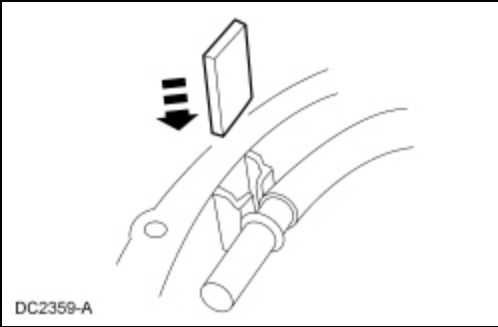


15. **NOTE:** Make sure the oil pump finger is in the slot in the case.

Install the oil pump assembly.

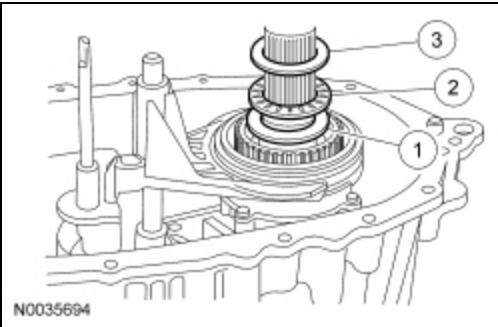


16. Install the oil pan magnet into the slot in the case.

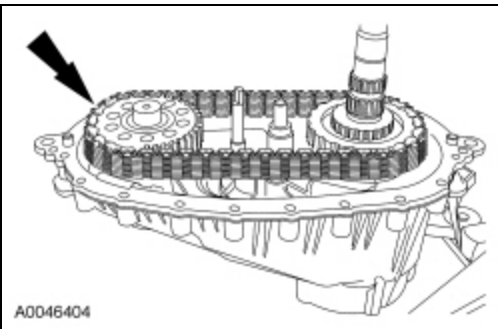


17. Install the 3-piece thrust bearing assembly.

1. Install the thrust washer.
2. Install the thrust bearing.
3. Install the thrust washer.



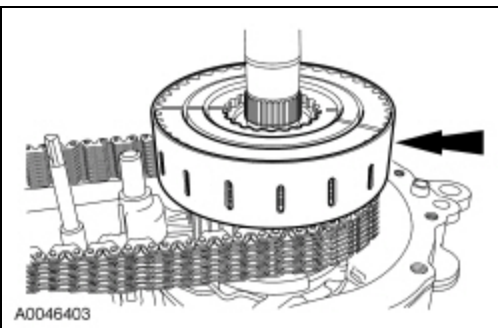
18. Install the drive chain and the sprockets as an assembly.



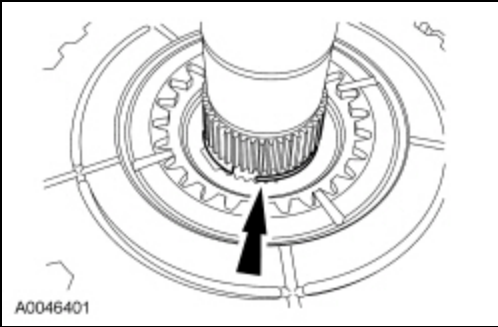
19. **NOTICE:** When installing the clutch assembly, do not separate the clutch pack assembly. The thrust washer in the lower clutch pack uses tabs to hold it in place. If the thrust washer is not in place, a transfer case clearance problem will occur.

NOTICE: If the clutch pack is disassembled, a new clutch pack must be installed and the armature transferred. Failure to install a new clutch pack can result in component failure.

Install the clutch pack assembly with the armature.

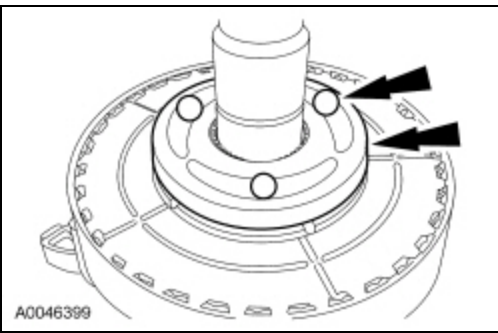


20. Install a new snap ring.

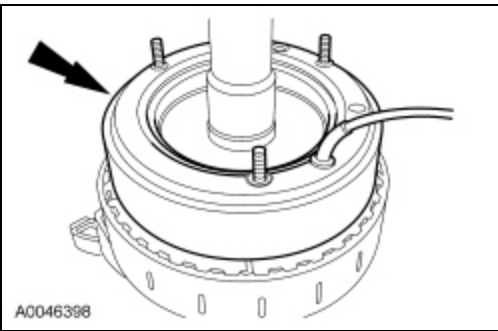


21. Install the wave spring.

22. Install the cam and the steel balls.



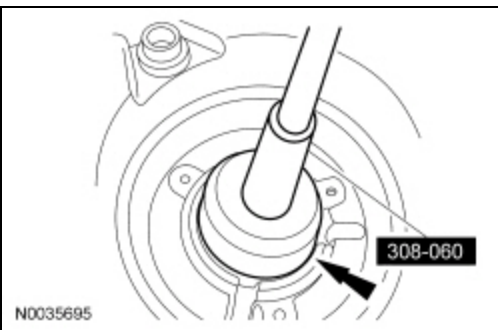
23. Install the cam/coil housing.



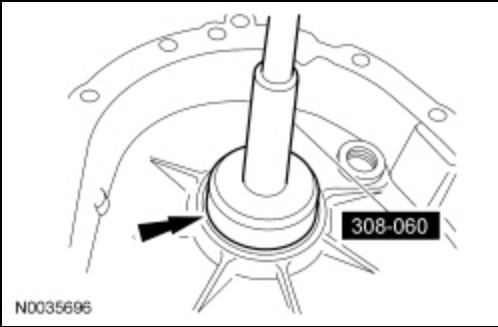
24. Install the thrust bearing assembly.

25. Install the snap ring.

26. Using a press and the Mainshaft Bearing Installer, install the rear output shaft support bearing.



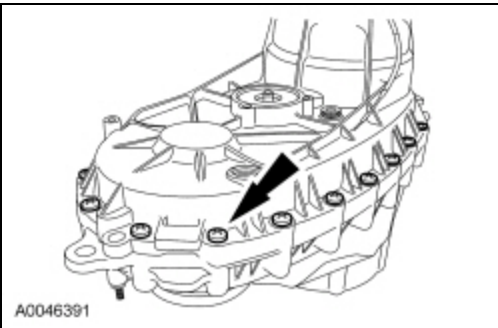
27. Using a press and the Mainshaft Bearing Installer, install the front output shaft rear bearing.



28. **NOTE:** Only apply enough silicone sealant to seal the transfer case halves together.

Apply a bead of silicone sealant to the transfer case mating surface, then position the 2 transfer case halves together. Tighten the 20 transfer case bolts evenly in a star pattern.

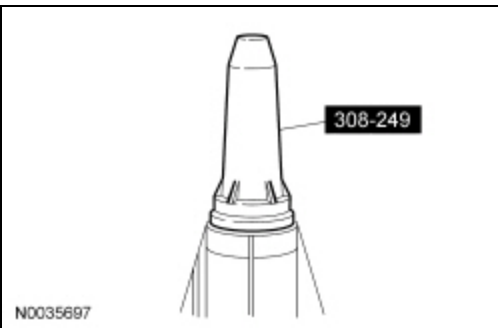
- Position the coil studs through the case holes, making sure the shift cam shaft is aligned with its case bore and the lockup shift fork spring is in position.
- Tighten to 29 Nm (21 lb-ft).



29. Install the 3 coil washers and nuts.

- Tighten to 10 Nm (89 lb-in).

30. Using the Input Shaft Oil Seal Installer, install the rear output shaft seal.



31. Apply a small bead of silicone sealant to the transfer case shift motor mating surface.

NOTICE: When installing the shift motor, tighten the bracket nuts last. Failure to follow the steps in the correct order may result in damage to the component.

32. Install the transfer case shift motor and the 3 bolts.

- Tighten to 10 Nm (89 lb-in).

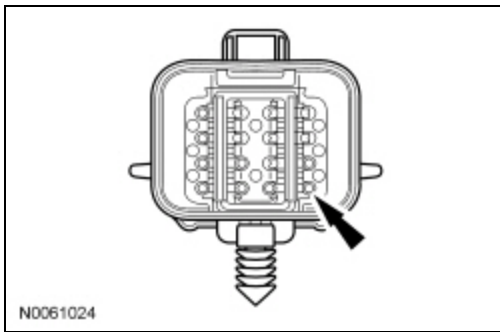
33. Install the transfer case shift motor bracket bolt.

- Tighten to 10 Nm (89 lb-in).

34. Secure the transfer case shift motor bracket nuts.

35. **NOTE:** After the wire is inserted into the connector, gently pull the wire back to verify the pin is locked inside the connector.

Insert the coil wire pin in the electrical connector and install the inner connector retainer.



36. Secure the transfer case to a transmission jack and remove the transfer case from the Transmission Holding Fixture.

